

An Adaptive Graph-Fusion Extension to Conditional Local Convolution for Short-Range WeatherBench Forecasting

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ABSTRACT

Conditional Local Convolution Recurrent Network (CLCRN) is an established graph-recurrent model for meteorological forecasting. We examine whether a compact adaptive graph-fusion (AGF) encoder can improve CLCRN by learning non-local node dependencies before the original recurrent backbone. The extension constructs a directed top- k graph from trainable source and destination node embeddings and combines graph-mixed and residual features through a learned gate. Experiments use the WeatherBench preprocessing and four tasks from the original CLCRN study: 2-m temperature, relative humidity, 10-m surface wind, and total cloud cover. We compare CLCRN-AGF with a schedule-matched CLCRN control over five seed replicates. The clearest effect is relative humidity: CLCRN-AGF reduces mean absolute error by 2.28% and root mean squared error by 3.02%; all seeds improve (Welch tests: $p = 7.6 \times 10^{-4}$ and $p = 8.4 \times 10^{-5}$). Effects on temperature, wind, and cloud cover are small or metric-dependent. Component ablations show that the adaptive graph branch contributes most clearly for humidity, whereas replacing the learned gate by mean fusion remains competitive. These results narrow the contribution of adaptive graph fusion: it is useful for the humidity task but is not a uniformly beneficial replacement for the original CLCRN encoder.

1 Introduction

Numerical weather prediction remains the operational standard because it encodes atmospheric physics¹, while data-driven models provide increasingly capable alternatives for selected forecasting tasks. WeatherBench established a common ERA5-based benchmark for evaluating such models². At the large-model end of the spectrum, FourCastNet, Pangu-Weather, and GraphCast demonstrate the potential of neural forecasting at global scale^{3–5}. Smaller graph-recurrent systems remain relevant when retraining cost, task-specific adaptation, and controlled experiments are important.

Conditional Local Convolution Recurrent Network (CLCRN) was introduced by Lin et al.⁶ for meteorological forecasting on an unstructured spherical grid. Its Conditional Local Convolution (CLConv) generates location-dependent kernels from spherical coordinates, geodesic distance, and directional information. The published model therefore already addresses the spatially shared-kernel limitation of conventional graph convolutions. The present study does not claim CLCRN or CLConv as new. Instead, it reproduces the public implementation and evaluates a specific extension motivated by adaptive graph learning.

The extension, termed **CLCRN-AGF**, adds an adaptive graph-fusion encoder before the original CLCRN recurrent backbone. Trainable source and destination node embeddings define a directed similarity matrix. A sparse top- k neighbourhood is used to aggregate node features, and a learned gate mixes the graph-aggregated representation with a residual projection. This design tests a focused hypothesis: non-local learned dependencies may add value beyond CLCRN’s geographically conditioned local convolution, particularly for variables whose dynamics are not fully described by local proximity.

The study is framed as an extension and reproducibility analysis. We compare CLCRN-AGF against a schedule-matched control that disables the complete AGF module while retaining the same data, optimizer, training budget, and random seeds. Published CLCRN values and an independent reproduction are reported only as contextual references. The contributions are:

- an adaptive top- k graph-fusion encoder attached to the established CLCRN backbone;
- a five-replicate comparison against a schedule-matched control, including confidence intervals and Welch tests;
- component ablations separating the adaptive graph branch from the learned fusion gate;
- a transparent separation between published reference values, independent reproduction, and newly generated results; and

- a negative as well as positive result: the extension consistently helps relative humidity but gives small or metric-dependent changes for the other tasks.

Related work

Conditional local convolution

CLCRN combines a sequence-to-sequence recurrent architecture with CLConv, whose kernel is conditioned on the coordinates of each centre-neighbour pair⁶. The original paper evaluated temperature, relative humidity, surface wind, and cloud cover on a 2,048-node spherical grid. CLConv and the CLCRN recurrent cell are prior work and serve as the fixed backbone in the present study.

Adaptive graph forecasting

Adaptive graph learning is commonly implemented with node embeddings, attention scores, or learned adjacency matrices. AGCRN learns node-specific patterns from adaptive embeddings⁷; Graph WaveNet and related models infer latent spatial dependencies alongside temporal forecasting^{8,9}; and adaptive spatio-temporal transformers revise fixed graph structure with attention^{10,11}. These methods motivate the question addressed here: whether a non-local adaptive branch adds useful information to a backbone that already contains geometry-conditioned local kernels.

Compact weather models

The experiments follow the original CLCRN task definition rather than the full-state, medium-range setting used by recent global forecasters. Related work has explored cost-efficient knowledge-oriented weather models, spherical neural operators, and learned post-processing of numerical forecasts, and ERA5-based microclimate prediction^{12–16}. Their datasets and objectives differ from the single-variable WeatherBench tasks considered here, but they illustrate the continuing role of compact, task-specific forecasting systems.

Results

Primary replicate comparison

The primary comparison uses five random-seed replicates {2021, 2022, 2023, 2024, 2025}. The control is the original CLCRN backbone trained with the same revised schedule as CLCRN-AGF. Table 1 reports mean and standard deviation across seeds, the mean difference (CLCRN-AGF minus control), a 95% Welch confidence interval for that difference, and a two-sided Welch test. The same seed labels are used in both queues, but the AGF module changes model initialization order; the analysis is therefore treated as a replicate comparison rather than a strict common-random-number paired design.

Table 1. Five-replicate comparison of CLCRN-AGF and the schedule-matched CLCRN control. Negative differences favour CLCRN-AGF. Confidence intervals and p -values use Welch’s unequal-variance approximation.

Task	Metric	Control ($\mu \pm \sigma$)	CLCRN-AGF ($\mu \pm \sigma$)	Difference [95% CI]	p_{Welch}
Temperature	MAE	1.229 ± 0.057	1.233 ± 0.032	$+0.004 [-0.067, +0.075]$	0.8985
	RMSE	2.097 ± 0.073	2.020 ± 0.047	$-0.076 [-0.169, +0.016]$	0.0921
Relative humidity	MAE	4.633 ± 0.032	4.528 ± 0.013	$-0.106 [-0.144, -0.067]$	7.6×10^{-4}
	RMSE	7.237 ± 0.051	7.019 ± 0.040	$-0.218 [-0.285, -0.151]$	8.4×10^{-5}
Surface wind	MAE	1.244 ± 0.005	1.240 ± 0.007	$-0.004 [-0.014, +0.006]$	0.3495
	RMSE	1.977 ± 0.008	1.973 ± 0.017	$-0.004 [-0.024, +0.017]$	0.6851
Cloud cover	MAE	0.1487 ± 0.0007	0.1480 ± 0.0013	$-0.0007 [-0.0023, +0.0009]$	0.3106
	RMSE	0.2473 ± 0.0019	0.2470 ± 0.0027	$-0.0003 [-0.0038, +0.0031]$	0.8368

Relative humidity is the only task for which every seed label favours CLCRN-AGF for both metrics. Its mean MAE decreases by 2.28% and RMSE by 3.02%. Temperature shows a metric-dependent result: MAE is essentially unchanged, whereas mean RMSE decreases by 3.64% with an interval that includes zero. Wind and cloud-cover changes are small relative to seed variability.

Contextual benchmarks

Table 2 distinguishes results generated in this study from external published values. Persistence is evaluated directly on the current test split. “Published CLCRN” is copied from Lin et al.⁶ after restoring the humidity and cloud-cover reporting scales

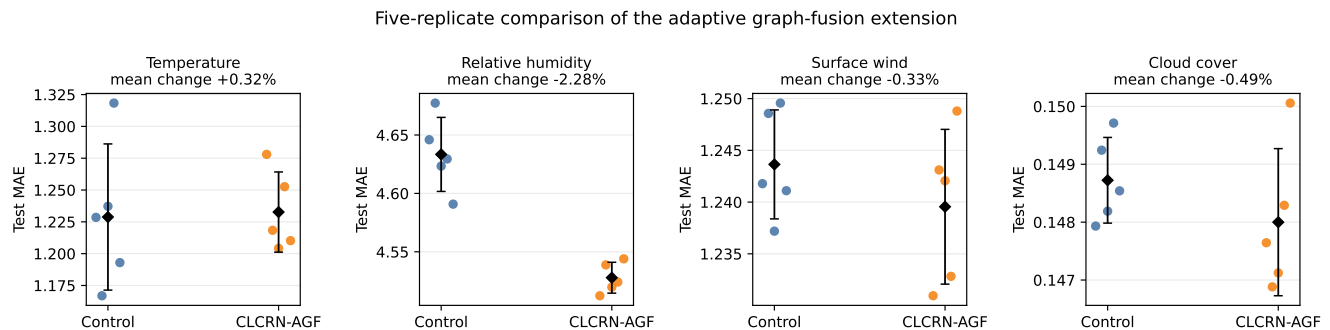


Figure 1. Test MAE for the schedule-matched control and CLCRN-AGF. Dots show individual seed replicates; diamonds show mean $\pm$ one standard deviation. The humidity improvement is consistent across all five seed labels, whereas the other tasks show overlapping or mixed changes.

used by the current evaluation code. “Reproduced CLCRN” is a single independent run of the public implementation. The two final rows are five-seed results generated under the revised schedule.

Table 2. Contextual test results. Published CLCRN values are external references, not newly generated results. Reproduced CLCRN is a single run; control and CLCRN-AGF are five-seed means. Lower is better.

Method	Temperature		Relative humidity		Surface wind		Cloud cover	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
Persistence (current split)	1.421	2.700	6.425	10.447	1.537	2.535	0.1620	0.2725
Published CLCRN ⁶	1.169	1.883	4.531	7.078	1.326	2.129	0.1491	0.2456
Reproduced CLCRN (one seed)	1.192	2.246	4.677	7.309	1.384	2.266	0.1564	0.2574
Schedule-matched control (five seeds)	1.229	2.097	4.633	7.237	1.244	1.977	0.1487	0.2473
CLCRN-AGF (five seeds)	1.233	2.020	4.528	7.019	1.240	1.973	0.1480	0.2470

The contextual comparison shows why the extension should not be described as uniformly superior. CLCRN-AGF improves on the published reference for relative humidity and wind, is mixed for cloud cover, and does not improve temperature. Differences from the external reference also combine architecture, software environment, and training-protocol effects; the schedule-matched control in Table 1 is therefore the primary causal comparison.

Component ablations

The experiment archive also contains component ablations. Table 3 separates three kinds of evidence: 100-epoch, three-seed ablations for temperature and relative humidity, and a 20-epoch single-seed screening run over all four tasks. The full-budget humidity comparison uses seeds 2023–2025. The w/o gated fusion humidity completion was trained with automatic mixed precision after repeated standard-precision runs exhausted the local CUDA allocator; checkpoint selection and test evaluation still use the same corrected zero-inclusive evaluator.

In the 100-epoch temperature ablation, removing the entire adaptive graph branch worsens RMSE relative to the full model, while replacing the learned gate by mean fusion gives similar or slightly lower mean errors. Thus the gate is not uniformly beneficial for temperature. In the full-budget humidity ablation, both adaptive-graph variants improve on the model without the adaptive graph branch. The full gated model has the lowest mean MAE and RMSE, but the mean-fusion variant remains close and was completed under the mixed-precision caveat described above. We therefore treat this block as sensitivity evidence that supports adaptive graph mixing, not as a definitive precision-matched test of the gate. Figure 2 visualizes the two full-budget ablation blocks with seed-level points. The 20-epoch screening run reaches the same qualitative warning: relative humidity is the most sensitive task, whereas wind, temperature, and cloud-cover screening results are mixed.

Computational cost

The AGF encoder approximately doubles the number of trainable parameters and increases inference memory because it constructs a dense node-similarity matrix before top- k selection. Table 4 reports direct measurements on the same NVIDIA GeForce RTX 5090 Laptop GPU. The extension is compact relative to global foundation models, but it is not cost-free and is not asymptotically linear in the number of nodes.

Table 3. Component ablations. The first two blocks report 100-epoch three-seed ablations at the full 12-hour horizon; the final block reports a 20-epoch single-seed screening ablation for all four tasks. Lower is better.

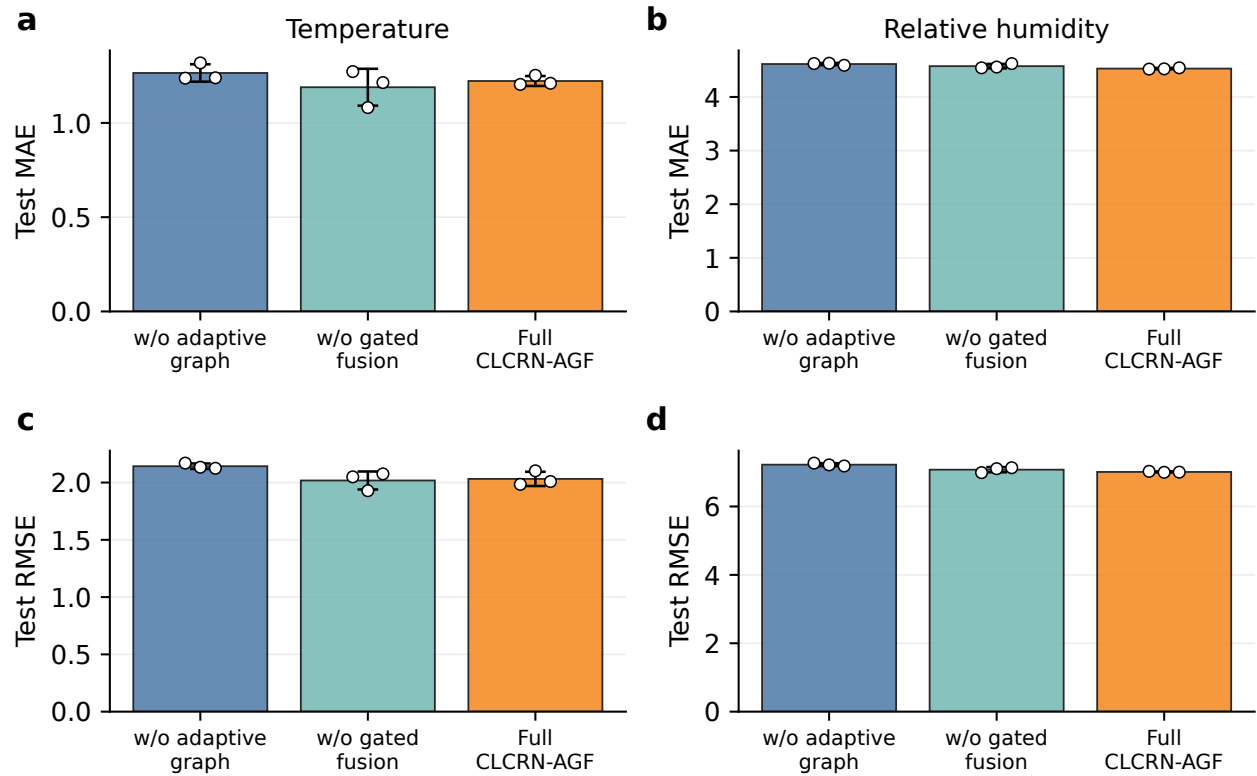
Budget	Task	Variant	MAE	RMSE
100 epochs, 3 seeds	Temperature	w/o adaptive graph	1.265 ± 0.046	2.143 ± 0.024
		w/o gated fusion	1.189 ± 0.098	2.018 ± 0.079
		Full CLCRN-AGF	1.222 ± 0.026	2.032 ± 0.062
100 epochs, 3 seeds	Relative humidity	w/o adaptive graph	4.615 ± 0.021	7.223 ± 0.043
		w/o gated fusion	4.575 ± 0.042	7.077 ± 0.076
		Full CLCRN-AGF	4.529 ± 0.013	7.011 ± 0.015
20 epochs, 1 seed	Temperature	w/o adaptive graph	2.105	3.193
		w/o gated fusion	2.716	3.665
		Full CLCRN-AGF	2.356	3.273
	Relative humidity	w/o adaptive graph	13.390	18.739
		w/o gated fusion	5.356	8.641
		Full CLCRN-AGF	5.151	8.097
	Surface wind	w/o adaptive graph	1.371	2.199
		w/o gated fusion	1.542	2.478
		Full CLCRN-AGF	1.483	2.319
	Cloud cover	w/o adaptive graph	0.1550	0.2511
		w/o gated fusion	0.1557	0.2551
		Full CLCRN-AGF	0.1559	0.2529

Values are mean $\pm$ sample standard deviation for the three-seed blocks. The humidity w/o gated fusion run used automatic mixed precision for training; all reported test metrics use the same zero-inclusive evaluator.

Table 4. Measured model size and inference cost. Timings are milliseconds per test batch; memory is peak allocated GPU memory during inference.

Task	Variant	Parameters	Inference (ms)	Memory (MB)
Temperature	CLCRN	39,268	202.6	352.3
	CLCRN-AGF	80,892	211.9	634.5
Relative humidity	CLCRN	39,268	206.7	353.1
	CLCRN-AGF	80,892	212.4	634.5
Surface wind	CLCRN	39,437	197.3	368.6
	CLCRN-AGF	81,061	207.4	644.6
Cloud cover	CLCRN	39,268	202.0	353.1
	CLCRN-AGF	80,892	203.4	634.5

Full-budget component ablations over three seed replicates



Bars show mean $\pm$ sample standard deviation; dots show individual seeds. Lower is better.

Figure 2. Full-budget component ablations for temperature and relative humidity. Bars show mean $\pm$ sample standard deviation across seeds 2023–2025, and dots show individual seed results. Panels a–b show test MAE; panels c–d show test RMSE. The relative-humidity w/o gated fusion run used automatic mixed precision during training; all reported test metrics use the same zero-inclusive evaluator. Lower is better.

Gate diagnostics

Gate values are summarized only as model diagnostics; their marginal distribution does not by itself establish a physical interpretation. The single-seed distributions in Figure 3 differ across tasks, but causal attribution would require interventions on the two branches or independent meteorological labels.

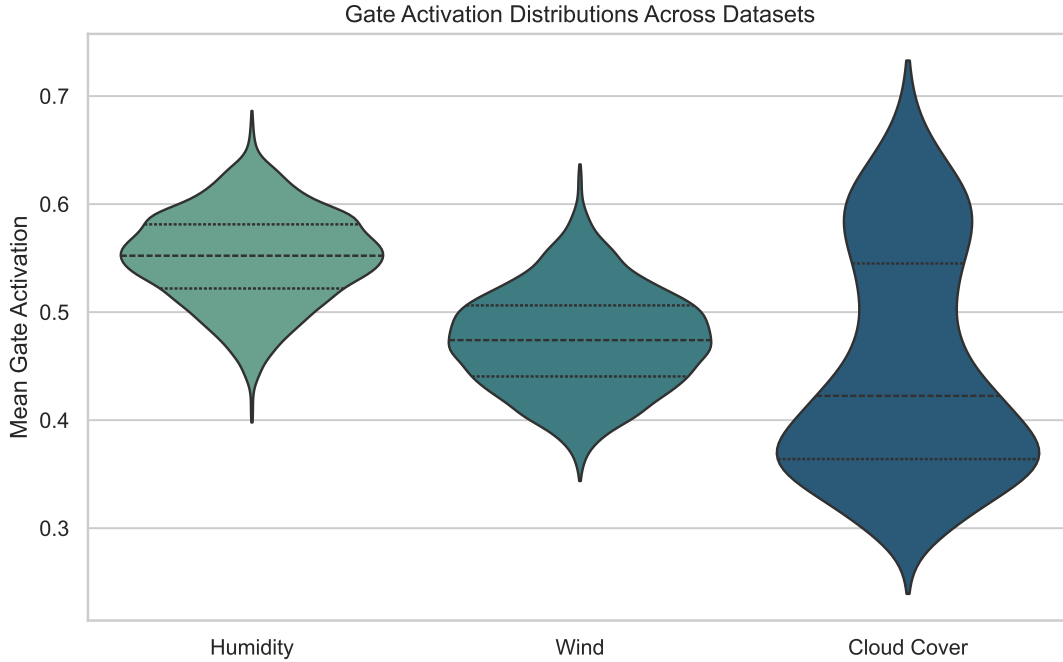


Figure 3. Distribution of mean gate activations for three tasks in a representative trained model. The plot describes how the learned mixture varies across samples. It should not be interpreted as proof that either branch corresponds to a specific physical process.

Discussion

The replicate experiment supports a narrow conclusion. Adding adaptive graph fusion to CLCRN is beneficial for relative humidity under the tested setup: the improvement is consistent across all five seed labels and the Welch confidence interval excludes zero for both MAE and RMSE. For temperature, wind, and cloud cover, the evidence does not support a general claim of architectural superiority.

This task dependence is plausible but should not be over-interpreted. Relative humidity may benefit from dependencies that are not captured by the fixed local neighbourhood alone. The current experiment does not demonstrate that the learned edges correspond to teleconnections or moisture corridors. Establishing such a claim would require spatial edge analysis, comparisons with meteorological indices, and perturbation tests.

The study also separates optimization effects from architectural effects. The revised schedule substantially improves wind and cloud-cover results even without AGF, while the additional AGF gain is small. Reporting only the best single run would obscure this distinction. The five-replicate design is therefore central to the present conclusions. The ablations further show that the two AGF components do not contribute uniformly. For humidity, removing the adaptive graph branch gives the weakest full-budget result, whereas mean fusion without the learned gate remains close to the full model. For temperature, mean fusion can also match or outperform the learned gate. The evidence therefore supports the adaptive graph branch more strongly than it supports a universal benefit from gated fusion, although the humidity gate comparison should be interpreted cautiously because one ablation variant used mixed-precision training.

Several limitations remain. First, the study inherits the compact single-variable setup of the original CLCRN paper rather than evaluating full-state medium-range weather prediction. Second, the current baseline set contains persistence, the published CLCRN reference, and a schedule-matched control; modern graph and non-graph baselines still need to be retrained under the identical data and optimization pipeline. Third, the top- k graph is obtained from a dense $N \times N$ similarity matrix, giving quadratic intermediate cost. Fourth, full-budget component ablations are now available for temperature and relative humidity, but wind and cloud cover still rely on the shorter 20-epoch component screening and should not be over-interpreted. The

humidity mean-fusion completion used mixed-precision training, so a fully precision-matched ablation grid would be useful before making stronger claims about the gate.

Conclusion

This work evaluates an adaptive graph-fusion extension to the published CLCRN weather-forecasting model. The revised manuscript distinguishes prior CLCRN contributions, external published values, independent reproduction, and new experiments. In five-replicate tests, CLCRN-AGF consistently improves relative-humidity forecasting, while changes for temperature, wind, and cloud cover are small or metric-dependent. The results therefore support adaptive graph fusion as a task-dependent extension rather than a universally better replacement for the original CLCRN encoder.

Methods

Data and preprocessing

We use the same four WeatherBench-derived tasks and public preprocessing pipeline as the original CLCRN study⁶: 2-m temperature (K), relative humidity (percentage points), 10-m surface wind components (u and v , m s^{-1}), and total cloud cover (fraction). Hourly ERA5 records from 1 January 2010 through 31 December 2018 are mapped to a 32×64 latitude–longitude grid with $N = 2,048$ nodes.

Each example contains 12 consecutive hourly observations and targets the next 12 hours. Starting windows are sampled once per day (step size 24), producing 3,286 examples. The split is chronological and unshuffled: 2,300 training examples (70%), 329 validation examples (10%), and 657 test examples (20%). Input and target variables are standardized with means and standard deviations computed from the training subset only; metrics are calculated after inverse transformation.

CLCRN backbone

The backbone is the public CLCRN implementation of Lin et al.⁶. CLConv generates a local kernel from each centre-neighbour coordinate pair and multiplies it by geodesic-distance and orientation factors. The resulting operator replaces linear transforms in a graph-convolutional GRU and is used in a sequence-to-sequence encoder–decoder. We retain this backbone without claiming its architecture as a contribution of the present study.

Adaptive graph-fusion encoder

Let $\mathbf{F} \in \mathbb{R}^{B \times T \times N \times d}$ be the concatenation of the projected weather feature and the learned node embedding. Trainable source and destination embeddings $\mathbf{E}_s, \mathbf{E}_t \in \mathbb{R}^{N \times d_e}$ define

$$\mathbf{A} = \text{softmax}(\mathbf{E}_s \mathbf{E}_t^\top), \quad (1)$$

where softmax is applied row-wise. For each node, the implementation retains the self-edge and the $k - 1$ largest remaining entries, then renormalizes the selected weights. With neighbour set $\mathcal{T}_k(i)$,

$$\mathbf{m}_i = \sum_{j \in \mathcal{T}_k(i)} \tilde{A}_{ij} \mathbf{F}_j. \quad (2)$$

Graph-mixed and residual features are projected separately:

$$\mathbf{q}_i = \mathbf{W}_m \mathbf{m}_i + \mathbf{b}_m, \quad \mathbf{r}_i = \mathbf{W}_r \mathbf{F}_i + \mathbf{b}_r. \quad (3)$$

The gated output is

$$\mathbf{g}_i = \sigma(\mathbf{W}_g [\mathbf{r}_i \parallel \mathbf{q}_i] + \mathbf{b}_g), \quad (4)$$

$$\mathbf{z}_i = \mathbf{g}_i \odot \mathbf{q}_i + (1 - \mathbf{g}_i) \odot \mathbf{r}_i. \quad (5)$$

$\mathbf{z}$ is concatenated with the standard CLCRN encoder inputs before the unchanged recurrent encoder. The control sets the AGF encoder off, so it removes both adaptive mixing and its gate.

Training configuration

Both variants use two recurrent layers, 32 hidden units, an 8-dimensional weather-feature embedding, an 8-dimensional CLConv embedding, one CLConv propagation view, and curriculum learning in the autoregressive decoder. CLCRN-AGF uses 10-dimensional adaptive node embeddings, an 8-dimensional AGF output, and $k = 4$ selected neighbours including the self-edge. Training uses Adam with learning rate 0.01, batch size 32, gradient-norm clipping at 5, and a MultiStepLR schedule

with milestones $\{10, 20, 30, 40, 50\}$ and multiplicative factor 0.95. Models train for at most 100 epochs and the checkpoint with the lowest validation MAE is used for test evaluation.

Experiments were run with Python 3.13, PyTorch 2.10, CUDA 12.8, and an NVIDIA GeForce RTX 5090 Laptop GPU. Seeds 2021–2025 are applied to NumPy and all PyTorch random-number generators. We do not claim bitwise deterministic reproducibility across hardware or library versions. Primary control and CLCRN-AGF runs were trained in standard precision. The additional relative-humidity w/o gated fusion ablation in Table 3 was completed with PyTorch automatic mixed precision and batch size 32 because standard-precision attempts repeatedly failed with CUDA allocator exhaustion on the available Windows GPU.

Metrics and statistical analysis

MAE and RMSE are computed after inverse scaling. The corrected evaluator includes all finite target values; zero is not treated as missing because zero wind components and zero cloud cover are physically meaningful. Persistence repeats the final input observation over the complete forecast horizon. The primary analysis compares five seed replicates for each variant. For each metric we report the mean and sample standard deviation across seeds, the mean difference, a two-sided 95% Welch confidence interval, and a two-sided Welch test. All reported test rows use this zero-inclusive evaluator. For archived runs with existing checkpoints, validation selection was recomputed with the same corrected MAE among the saved checkpoints before test evaluation.

Computational complexity

The original CLConv implementation materializes dense $N \times N$ kernel operators, and AGF constructs a dense $N \times N$ similarity matrix before top- k selection. The implementation therefore has quadratic intermediate memory and similarity-computation cost in N , despite sparse aggregation after selection. Claims of linear scaling are not made.

Data availability

The raw ERA5 variables are available through the [WeatherBench GitHub repository](#) and the Copernicus Climate Data Store. The original CLCRN project also provides preprocessed WeatherBench files through the data link described in its public README. The preprocessing period, variables, temporal windows, and chronological split used here are specified in the Methods. Because the preprocessed arrays are derived from third-party ERA5/WeatherBench data, they are not redistributed in the article package. The submitted supplementary review archive contains the exact configuration files, corrected metric tables, per-seed summaries, and scripts needed to regenerate the reported tables and figures from the preprocessed files.

Code availability

The study uses the public CLCRN implementation, extended with the AGF module and the corrected evaluation scripts described in this manuscript. The source files, run configurations, corrected evaluation script, humidity component-ablation completion script, and per-seed result summaries are included in the supplementary software/review archive submitted with this manuscript. The same materials are deposited in a public GitHub repository ([gengx-prog/CLCRN-AGF-Scientific-Reports](#)), with the review archive attached to release [v0.1-review](#).

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213 Author contributions

214 X.G. conceived the extension, implemented the software, conducted the experiments and analysis, prepared the visualizations,

215 and wrote the original draft. H.-Y.L. supervised the study and reviewed and edited the manuscript. Both authors reviewed and

216 approved the final manuscript.

217 Competing interests

218 The authors declare no competing interests.

219 Generative AI and AI-assisted technologies disclosure

220 During preparation of this manuscript, the authors used AI-assisted language editing to improve readability and organization.

221 The authors reviewed and edited all generated content and take full responsibility for the final manuscript.